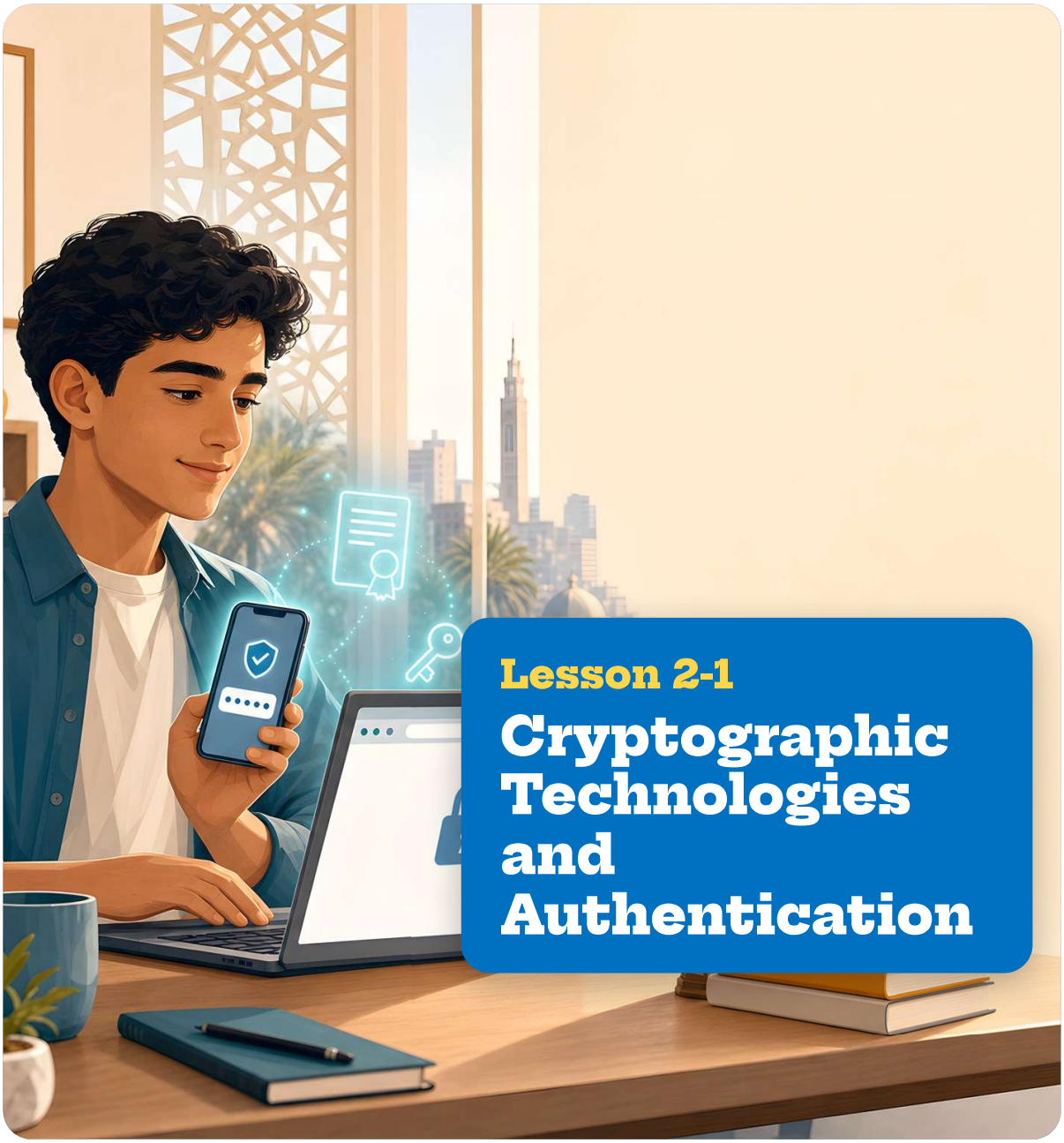


Lesson 2-1

Cryptographic Technologies and Authentication

Learning Objectives

- Explain how HTTPS secures communication: how the TLS handshake uses public-key cryptography to share a key safely, and how common-key cryptography then protects the data that follows.
- Describe how authentication, including multi-factor authentication, is used in real services, and explain why security technologies are combined.

Every time you log in to a bank app, buy something online, or open a website with a small padlock in the address bar, hidden technologies are working to keep your information safe. Messages are scrambled so no one else can read them, the website proves that it is genuine, and your identity is checked before you are granted access. Without these protections, passwords could be stolen and private data could be read by unauthorized people.

? Guiding question: How do cryptographic technologies and authentication work together to keep online communication and services secure?

Point!

1. How HTTPS Communication Works

(1) **(HTTPS (HTTP Secure))**…a communication protocol that adds TLS encryption to HTTP. By providing encryption, detection of tampering, and verification of the communication partner, it protects users from threats such as eavesdropping, tampering, and impersonation. The procedure for establishing a secure communication is called the TLS handshake.

(2) HTTPS combines common-key (symmetric-key) cryptography and public-key cryptography as follows, taking advantage of the strengths of both methods.

Stage	What happens	Cryptographic method used
① **Connection**	The web server sends a digital certificate and a public key to the browser. The browser verifies the server's authenticity through a certificate authority (CA).	Public-key cryptography
② **Key exchange**	The browser creates a common key for communication, encrypts it with the server's public key, and sends it. The server decrypts it with its private key to obtain the common key.	Public-key cryptography
③ **Data communication**	Data is encrypted and exchanged using the shared common key.	Common-key cryptography

Note: the exchange above shows one way a common key can be shared. Current versions of TLS use Diffie-Hellman key agreement instead.

Key Fact:
Secure online communication and services depend on combining cryptography and authentication — not on any single technology.

Key Concepts:

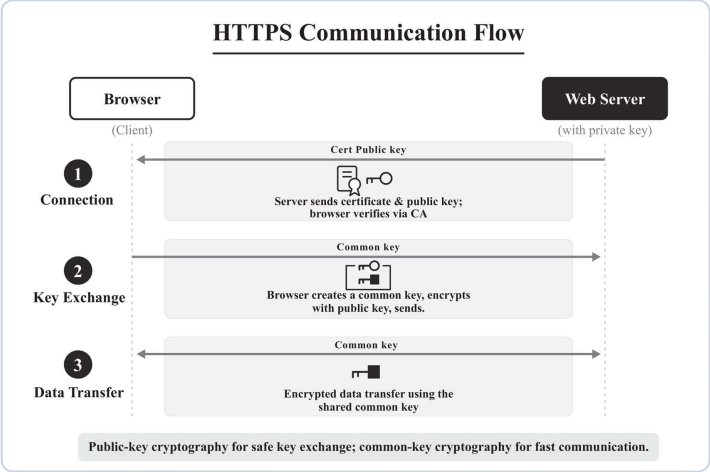
- HTTPS
- common-key cryptography
- public-key cryptography
- TLS handshake
- multi-factor authentication
- digital signature
- digital certificate
- non-repudiation

KEY TERMS

HTTPS — HTTP with TLS encryption added (encryption, tamper detection, and partner verification).



A public key locks; a different private key unlocks. In a digital signature the direction reverses: the private key signs and the public key verifies.



HTTPS Communication Flow

(3) Public-key cryptography is used to "share keys safely," and common-key cryptography is used to "exchange large amounts of data quickly." This division of roles makes it possible to achieve both security and communication speed.

2. Use of Authentication in Real Services

(1) Many services adopt **(multi-factor authentication)**, in which two or more of the three factors of authentication (knowledge, possession, biometric) are combined.

(2) When only one factor is used, an unauthorized party can gain access if that factor is broken. By combining multiple factors, even if one is broken, another can still prevent access.
e.g., If only a password (knowledge) is used, an unauthorized login is possible once the password is leaked. However, if a notification to a smartphone (possession) is also required, an attacker cannot log in without the device, even if the password is leaked.

(3) The following examples of multi-factor authentication are used in various services.

Service	Examples of authentication used
Online banking	Password (knowledge) + one-time password (possession)
SNS login	Password (knowledge) + SMS authentication (possession)
Company laptop login	Smart card (possession) + fingerprint authentication (biometric)
Online shopping	Password (knowledge) + one-time code from 3-D Secure (possession)

3. Combinations of Security Technologies

(1) Safe communication and services are achieved not by a single technology, but by combining multiple security technologies.



Strong login combines two or more different factors.

KEY TERMS
multi-factor authentication
— combining two or more of the three factors: knowledge, possession, biometric.

Technology	Effect
Encryption (common-key, public-key)	Prevents communication content from being read by third parties
Digital signature	Detects data tampering or impersonation of the sender, and makes it impossible for the sender to deny having sent the data afterward (non-repudiation)
Digital certificate	Verifies that the communication partner is authentic
Multi-factor authentication	Prevents unauthorized logins

(2) For example, when purchasing a product through online shopping, these technologies are combined and used as follows.

① When you access the site, HTTPS communication begins, and **(encryption)** prevents communication content from being read by third parties.

② The browser checks the **(digital certificate)** to confirm that the destination is the genuine shopping site.

③ For account login, **(multi-factor authentication)** combining a password and SMS authentication, etc., prevents unauthorized logins.

④ When sending order details, a **(digital signature)** can also be used so that tampering during communication can be detected.

(3) By combining multiple technologies in this way, various threats such as "eavesdropping," "tampering," "impersonation," and "unauthorized login" can be addressed simultaneously.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A HTTPS communication uses both public-key cryptography and common-key cryptography.

B HTTPS communication exchanges all data using public-key cryptography.

C Multi-factor authentication is a method that combines two or more of the three factors of authentication.

D An example of multi-factor authentication is setting two passwords.

(2) Match each threat (a–c) with the most appropriate security technology from the choices below (A–C).

a I want to prevent communication content from being read by third parties.

b I want to verify that data has not been altered during communication.

KEY TERMS
digital signature — detects tampering and impersonation, and provides non-repudiation.



A certificate authority vouches that a site is genuine.

💡 PAUSE & THINK

If a password is leaked, why can a second factor sent to a phone still keep the account safe?

KEY TERMS
public-key cryptography — shares a key safely.
common-key cryptography — exchanges data quickly.

c I want to verify that the communication partner is authentic.

[Choices] **A** Encryption **B** Digital signature **C** Digital certificate

✓ Solution

(1)

A HTTPS communication uses public-key cryptography for key exchange and common-key cryptography for data communication. Therefore, ○.

B Exchanging all data using public-key cryptography would be slow, so common-key cryptography is used for data communication. Therefore, ×.

C Correct definition of multi-factor authentication. Therefore, ○.

D Even if two passwords are set, both are "knowledge" factors, so this does not constitute multi-factor authentication (it is two-step authentication using the same factor). Therefore, ×.

(2)

a: Preventing reading of communication content is encryption. **A**

b: Detecting data tampering is a digital signature. **B**

c: Verifying the authenticity of the communication partner is a digital certificate. **C**

Try

1 Answer the following questions.

1. In HTTPS communication, what cryptographic method is used so that the browser and server can safely share a common key?
2. In HTTPS communication, what cryptographic method is used for data communication after the common key has been shared?
3. Briefly explain the reason for combining the cryptographic methods in (1) and (2), based on the characteristics of each method.

2 Answer the following questions.

(1) For each service (a–d), indicate which combination of authentication factors is used: **A** Knowledge + possession, **B** Knowledge + biometric, **C** Biometric + possession.

a Password + one-time password

b Password + fingerprint authentication

EXAM-STYLE QUESTION

Explain why HTTPS uses public-key cryptography to share the key but common-key cryptography to exchange the data. **[6] Refer to:** security and speed.

c Password + SMS authentication

d Face recognition + smart card

(2) From the following options (A–D), choose the one that is NOT an appropriate description of combinations of security technologies.

A Encryption is a technology that prevents communication content from being read by third parties.

B Digital signatures are a technology that improves communication speed.

C Digital certificates are a system for verifying that the communication partner is authentic.

D By combining multiple security technologies, various threats can be addressed.

Think as an Engineer — Investigate & Decide

A school in your city is building an online portal where students check their grades and pay fees. You must choose how students log in. Which is the better choice for this service?

☐ **A password only** ☐ **A password plus a one-time password sent to the phone**

1 · Collect data. Ask ten classmates which login methods they use and trust (password, one-time password, SMS, fingerprint or face). Record the results in a table and find the most common answer.

2 · Analyze the stakeholders. For each group — a student, the school, and a student with no smartphone — note one benefit and one drawback of requiring a second factor.

3 · Decide. Recommend one login method for the portal and give two reasons based on your survey and your table. **In pairs,** compare your recommendations and agree together on the single strongest reason before you write it down.

Give evidence for your answer.

Stuck? Start with the user: a password alone fails once it leaks; adding a possession factor (a one-time password) still blocks the login.

🌍 In a New Context

A small online shop wants to secure both customer login and payment. Using what you learned, name two security technologies it should use and say where each one helps, then identify one threat that could still remain.

❓ Lesson Question — Answered

The lesson asked how cryptographic technologies and authentication keep communication and services secure. HTTPS combines public-key cryptography — used to share a key safely — with common-key cryptography — used to exchange data quickly — so communication is both secure and fast. Beyond communication, services stay safe by combining several technologies: encryption hides content, digital certificates prove a site is genuine, digital signatures detect tampering, and multi-factor authentication blocks unauthorized logins. Security online therefore comes from layering these technologies so that many threats are addressed at the same time.

KEY TERMS
digital certificate — proves a site is genuine.
multi-factor authentication — blocks unauthorized logins.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

HTTPS communication, used to access websites safely, combines multiple cryptographic technologies. First, the web server sends a (a) and a public key to the browser. The browser verifies whether the server is authentic through (b). Next, the browser creates a (c) for communication and encrypts it with the server's public key before sending. After this, all data communication is encrypted using this (c).

1. Fill in the blanks (a)–(c).
2. Briefly explain the reason for using common-key cryptography rather than public-key cryptography for data communication.

⭐ REFLECT & CHALLENGE

Reflect: Which of these protections do you rely on most when you go online?
Challenge: Suggest one extra security measure an online shop could add, and one threat it would still not remove.

KEY TERMS
encryption — keeps content private.
digital signature — detects tampering; gives non-repudiation.

3 Answer the following questions.

(1) For each threat (a–d), indicate which technology is primarily used to address it: **A** Encryption, **B** Digital signature, **C** Digital certificate, **D** Multi-factor authentication.

a A password is leaked and unauthorized login occurs.

b Data is altered during communication.

c A user is led to a fake website.

d Communication content is intercepted by a third party.

(2) From the following options (A–D), choose the one that is NOT an appropriate example of multi-factor authentication.

A Password + smartphone notification

B Password + secret question

C Fingerprint authentication + smart card

D Password + SMS authentication

⭐ Key takeaway

Remember:
HTTPS uses public-key cryptography to share a key safely and common-key cryptography to move data quickly. Services stay safe by layering encryption, certificates, signatures, and multi-factor authentication.